Experiment No. 3 Implementation and Analysis of Spatial Filtering Techniques using Low-Pass and High-Pass Filters

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```
# Spatial Filtering in Python
import cv2
import numpy as np
import matplotlib.pyplot as plt
from google.colab import files

# Step 1: Upload an image
uploaded = files.upload()
image_path = list(uploaded.keys())[0]
```



Step 2: Read image in grayscale

```
img = cv2.imread(image_path, cv2.IMREAD_GRAYSCALE)
```



Step 3: Add synthetic Salt-and-Pepper Noise

```
noisy_img = img.copy()

num_noise = 500

coords = [np.random.randint(0, i - 1, num_noise) for i in
img.shape]

noisy_img[tuple(coords)] = 255   # Salt noise

coords = [np.random.randint(0, i - 1, num_noise) for i in
img.shape]

noisy_img[tuple(coords)] = 0     # Pepper noise
```



Step 4: Gaussian Blur (Low-Pass Filter)

```
gaussian_blur = cv2.GaussianBlur(noisy_img, (5, 5), 0)
```

Gaussian Blur



Step 5: Median Filter (Salt-and-Pepper noise removal)

```
median_filter = cv2.medianBlur(noisy_img, 5)
```

Median Filter



Step 6: Average (Mean) Filter

```
average_filter = cv2.blur(noisy_img, (5, 5))
```



Step 7: Laplacian Filter (High-Pass Filter - Edge Enhancement)

```
laplacian = cv2.Laplacian(img, cv2.CV_64F)
```

```
laplacian = np.uint8(np.absolute(laplacian))
```



Step 8: Sobel Edge Detection (Horizontal & Vertical)

```
sobel_x = cv2.Sobel(img, cv2.CV_64F, 1, 0, ksize=3) # Horizontal edges
```

```
sobel_y = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3) # Vertical edges
```

```
sobel_combined = cv2.magnitude(sobel_x, sobel_y)
```

```
sobel_combined = np.uint8(np.absolute(sobel_combined))
```

Sobel X (Horizontal)



Sobel Y (Vertical)



Sobel Combined



```
# Step 9: Display results
titles = [
    'Original Image', 'Noisy Image',
    'Gaussian Blur', 'Median Filter',
    'Average Filter', 'Laplacian Edges',
    'Sobel X (Horizontal)', 'Sobel Y (Vertical)',
    'Sobel Combined'
]
images = [
    img, noisy_img,
    gaussian_blur, median_filter,
    average_filter, laplacian,
    sobel_x, sobel_y,
    sobel_combined
]
plt.figure(figsize=(12, 10))
for i in range(9):
    plt.subplot(3, 3, i + 1)
    plt.imshow(images[i], cmap='gray')
    plt.title(titles[i])
    plt.axis('off')
plt.tight_layout()
plt.show()
```

1. What is spatial filtering? How is it used in digital image processing?

ANS - Spatial filtering is the process of modifying pixel values using their neighbors with a kernel. It is used in digital image processing for noise reduction, smoothing, sharpening, and edge detection, making images clearer and features more detectable.

2. Differentiate between Low-Pass Filters and High-Pass Filters with suitable examples.

ANS - Low-Pass Filters

- Pass low frequencies, block high frequencies.
- Effect: Smooths image, reduces noise, edges blur.
- Example: Gaussian Blur, Median Filter.

High-Pass Filters

- Pass high frequencies, block low frequencies.
- Effect: Sharpens image, highlights edges.
- Example: Laplacian, Sobel Operator.

3. Compare Average Filter, Gaussian Filter, and Median Filter based on their working principles and applications.

ANS - Average Filter

- Principle: Replaces each pixel with the mean of its neighborhood.
- Effect: Smooths image, reduces random noise, but blurs edges.
- Application: Simple noise reduction, basic smoothing.

Gaussian Filter

- Principle: Weighted average using Gaussian distribution.
- Effect: Smooths image while preserving structure better than average filter.
- Application: Noise reduction in medical imaging, preprocessing in vision tasks.

Median Filter

- Principle: Replaces each pixel with the median of its neighborhood.
- Effect: Removes salt-and-pepper noise effectively, preserves edges.
- Application: Best for impulse noise removal in scanned documents or photos.

4. Why is the Median Filter particularly effective for removing salt-and-pepper noise?

ANS –

- Median Filter replaces each pixel with the median value of its neighborhood.
- Salt-and-pepper noise consists of extreme black or white pixels.
- Taking the median ignores these outliers and keeps the central tendency.
- Result: Noise is removed while edges remain sharp.

5. Explain the role of convolution kernels in spatial filtering.

ANS -

- Convolution Kernel is a small matrix (mask) applied to an image.
- It defines how pixel values are combined with their neighbors.
- By sliding the kernel across the image, different effects are produced:
- Low-pass kernels → smoothing, noise reduction.
- High-pass kernels → edge detection, sharpening.

Examples:

- Mean kernel → averaging filter.

6. What is the purpose of the Sobel and Laplacian operators in edge detection?

ANS - Sobel Operator

- Uses first-order derivatives to detect gradients in horizontal and vertical directions.
- Highlights strong edges and preserves orientation information.
- Commonly used for detecting object boundaries.

Laplacian Operator

- Uses second-order derivatives to detect rapid intensity changes.
- Highlights fine details and edges without orientation.
- Useful for sharpening and edge enhancement.

7. Why are filtering operations considered an essential preprocessing step in computer vision?

ANS -

- Filtering improves image quality before analysis.
- Removes noise and unwanted variations.
- Enhances edges and features for better detection.
- Ensures algorithms like segmentation, recognition, and tracking work more accurately.

8. Discuss the trade-off between image smoothing and edge preservation during filtering.

ANS -

- Smoothing reduces noise and makes the image clearer.
- But excessive smoothing blurs edges and important details.
- Edge Preservation keeps boundaries sharp for recognition.
- However, preserving edges may leave some noise unremoved.
- Trade-off: More smoothing → less noise but weaker edges; less smoothing → stronger edges but more noise.

9. Mention four real-world applications where spatial filtering techniques are widely used.

ANS –

- **Medical Imaging** → Noise reduction in CT/MRI scans for clearer diagnosis.
- **Remote Sensing** → Enhancing satellite images for land use and environmental studies.
- **Surveillance** → Edge detection for object tracking and motion analysis.
- **Autonomous Driving** → Preprocessing for lane detection and obstacle recognition

10. Compare spatial domain filtering with frequency domain filtering in terms of implementation and practical applications

ANS –

Spatial Domain Filtering

- Works directly on image pixels using kernels.
- Easy to implement, computationally simple.
- Applications: noise reduction, smoothing, edge detection (Gaussian, Median, Sobel).

Frequency Domain Filtering

- Works on Fourier transform of the image.
- More powerful for periodic noise removal and complex filtering.
- Applications: removing periodic noise, image compression, advanced enhancement.